the Earth rotates through them. If the Earth did not rotate, the squeeze and stretch would be constant, unchanging.

The second difference is the direction of the tides (Figures 10.3a,b): The Moon produces tidal forces in all spatial directions. It stretches the oceans in the *longitudinal* direction (toward and away from the Moon), and it squeezes the oceans in *transverse* directions (perpendicular to the Moon's direction). By contrast, a gravitational wave produces no tidal forces at all in the longitudinal direction (along the direction of the wave's propagation). However, in the transverse plane, the wave stretches the oceans in one direction (the up–down direction in Figure 10.3b) and squeezes along the other direction (the front–back direction in Figure 10.3b). This stretch and squeeze is oscillatory. As a crest of the wave passes, the stretch is up-down, the squeeze is front–back; as a trough of the wave passes, there is a reversal to up–down squeeze and front–back stretch; as the next crest arrives, there is a reversal again to up–down stretch and front–back squeeze.